

Appendix E - Brief User Guide

This appendix guides the reader through installing PHAROS and creating, running, and inspecting their own scenarios in the graphical application or with the standalone runner. The public repository provides the Windows release, source code, and technical documentation (Raiher, 2026b). Detailed descriptions of scenario files, command-line execution, and the controller interface are supplied in `Docs/Reference/TGSCN.md`, `Docs/GettingStarted/StandaloneRunner.md`, and `ControllerSDK/PHAROSControllerAPI.h`, respectively. The screenshots are unaltered full-frame captures of the PHAROS interface. They serve as visual references; the table also describes options that may lie below the visible scroll position.

E.1 Installation and User Data

PHAROS is a 64-bit Windows desktop application. Run the Windows installer and open PHAROS from the Start Menu or the optional desktop shortcut. The default installation directory is `%LOCALAPPDATA%\Programs\PHAROS`; `PHAROS.exe` can also be launched directly from that directory. Neither Unreal Engine nor Visual Studio is required to use the installed application or its standalone runner. On first launch, the application asks the user to review and accept its end-user license agreement.

The application creates user-data folders as needed under `%LOCALAPPDATA%\PHAROS\Saved`. The scenario library is stored in `SaveGames\TGSimulationLibrary.sav`, controller files in `Controllers`, and numerical results in `SimulationResults`. Section E.3 identifies the CSV files available for independent analysis. Scenario-library backups in `SaveGames\Backups` preserve earlier saved versions for restoration if saving fails. Files in `SaveGames\Recovery` preserve libraries that could not be opened, keeping the original data available for investigation or attempted recovery.

Launching PHAROS opens the main menu in Figure E.1. *New Scenario* begins a new scenario, *Scenario Library* opens saved scenarios and their available results, *About & Documentation* opens the bundled reference material, and *Exit* closes the application.



FIGURE E.1 – PHAROS main menu.

E.2 Scenario Authoring

The configuration screen edits a scenario, imports a .tgscn file, saves the current draft, or opens the Scenario Library. Importing a TGSCN automatically fills the configuration panels with the scenario’s settings. Table E.1 identifies the panels and their corresponding screenshots. Imported CSV files contain numerical values separated by commas, without header rows.

The panels may be edited in any order, subject to their input dependencies. The Component Tree supplies the components and joint DOFs referenced by the actuators, controller, solar radiation pressure settings, and articulated aerodynamic database. These entries therefore need to match the configured spacecraft. Aerodynamics also requires an enabled atmosphere. The Review panel checks the completed configuration before execution.

TABLE E.1 – Configuration panels and principal options in PHAROS.

Panel	Principal Options	Figure
Scenario and Solver	Scenario name; initial UTC; end by duration or final UTC; RK4 or Dormand–Prince 5(4); integration steps and tolerances; recording interval; numerical and stored-output limits.	E.2
Initial State	Reference frame; Cartesian position and velocity; scalar-first attitude quaternion; angular velocity.	E.3
Component Tree	Component hierarchy; masses, centers of mass, inertias, joint anchors and orientations; rotational and translational DOFs with their initial coordinates, rates, and limits; primitive or imported-STL geometry and visual appearance.	E.4

Continued on the next page.

TABLE E.1 – Configuration panels and principal options in PHAROS (continued).

Panel	Principal Options	Figure
Actuators	Prescribed or controller-commanded thrusters, their mounting and propellant components, application points, directions, firing windows, thrust, and specific impulse; reaction and momentum wheels, their axes, initial momenta, and limits.	E.5–E.6
Controls	No controller or a user C++ controller selected from the Controller Library. The library provides a source template, source import, compilation, and prebuilt-DLL import. The <i>Controller Reference</i> button opens usage instructions.	E.7–E.8
Gravity and Bodies	Bodies, gravity activation, activation and barycenter-resolution radii, and relativistic correction. A harmonic-gravity CSV has two values in its first row (model gravitational parameter and reference radius), then four per row (degree, order, normalized cosine and sine coefficients). The maximum degree selects the coefficients used.	E.9
Solar Radiation Pressure	Enablement, celestial eclipses, component shadowing, surface tessellation, and absorption, specular, and diffuse fractions, with component- and facet-specific optical properties.	E.10
Atmosphere	Central body and atmospheric model. A general-profile CSV needs at least two rows and five columns: altitude, density, temperature, mean particle mass, and collision cross section. Cubic Harris–Priester is available only for Earth; it uses the specified solar flux, a 50×8 density-coefficient CSV, and a molecular-profile CSV with at least two rows and four columns: altitude, temperature, mean particle mass, and collision cross section.	E.11
Aerodynamics	Reference area and length, dynamic-pressure limits, constant-drag coefficient, and database interpolation/extrapolation options. An imported database row has 11 columns plus one per joint DOF: molecular speed ratio, Knudsen number, three incoming-flow direction components, the joint coordinates, three force coefficients, and three moment coefficients.	E.12
Review	Configuration and input-file checks. Scenario configuration errors must be corrected; warnings require acknowledgement. Accepting the report enables <i>Simulate</i> ; changing the configuration requires a new review.	E.13

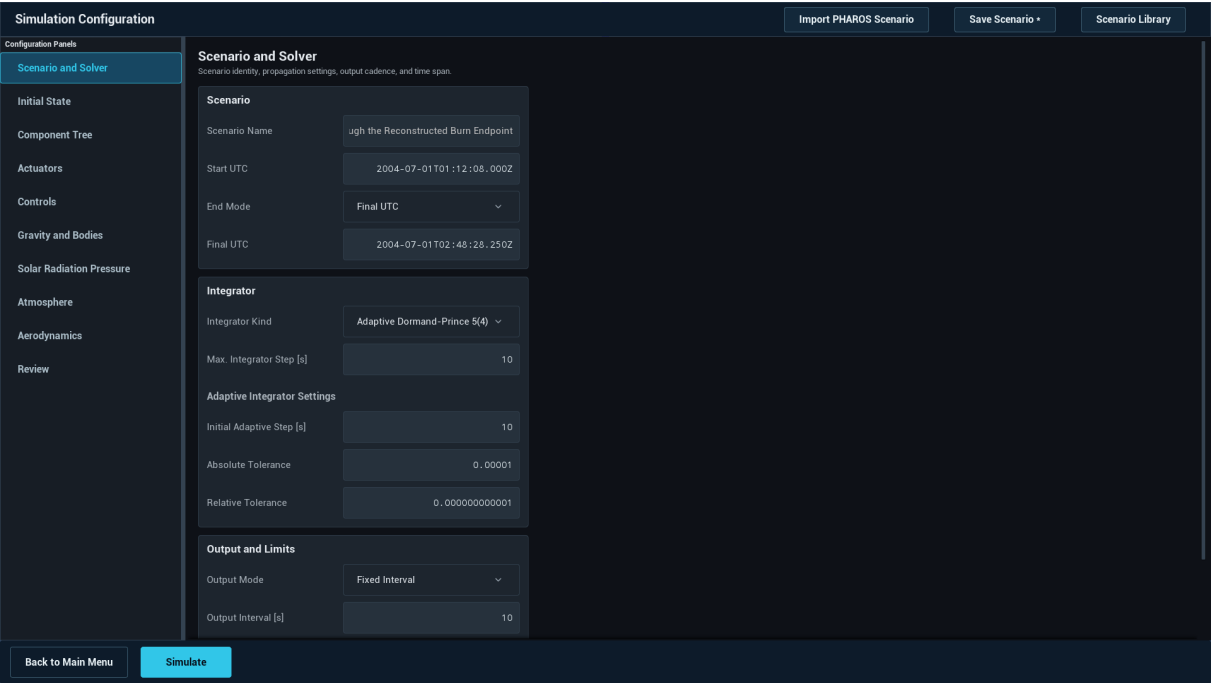


FIGURE E.2 – Scenario and Solver panel.

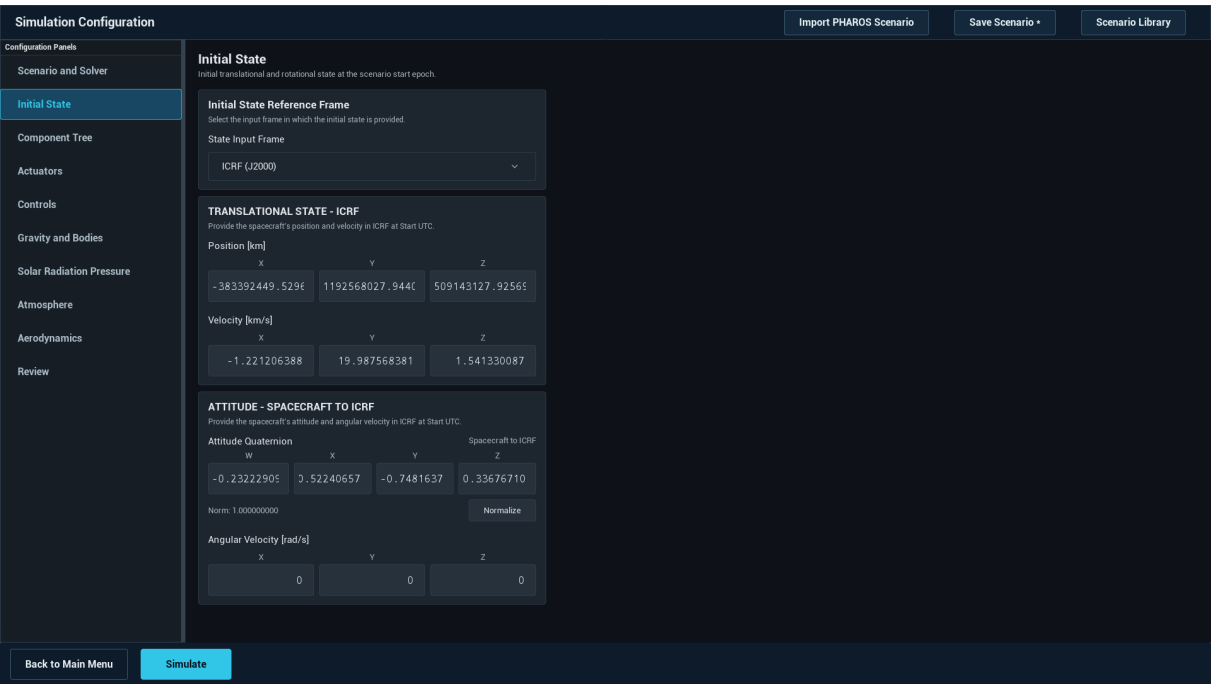


FIGURE E.3 – Initial State panel.

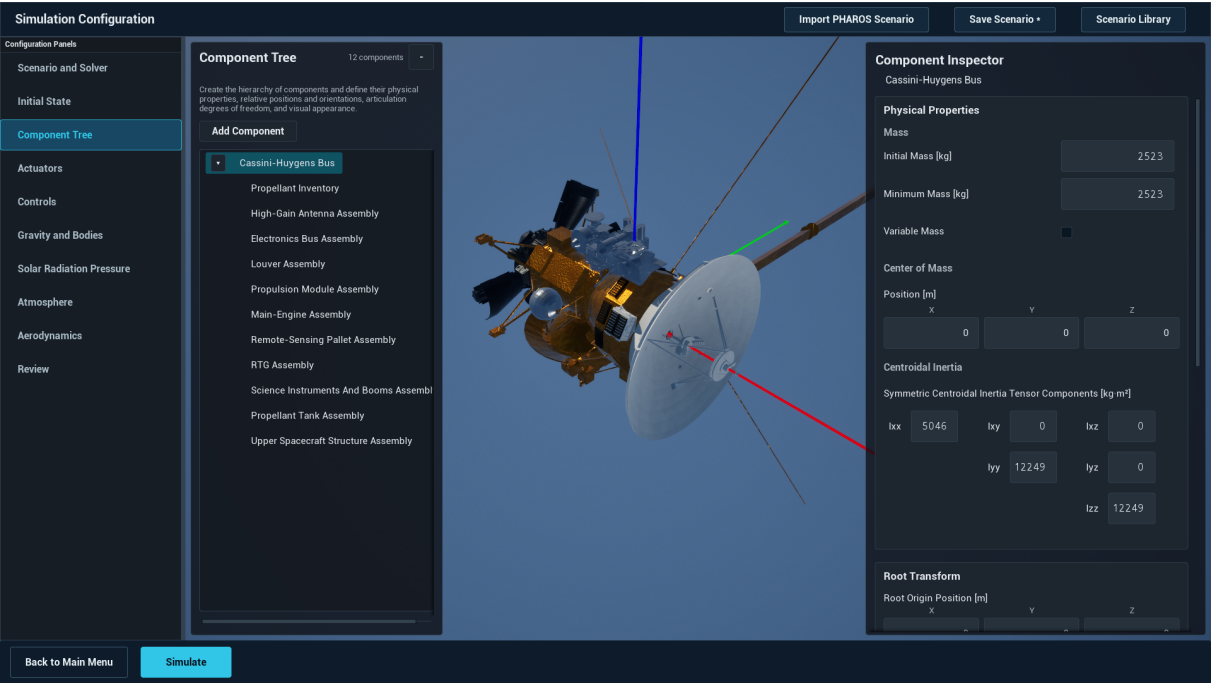


FIGURE E.4 – Component Tree panel.

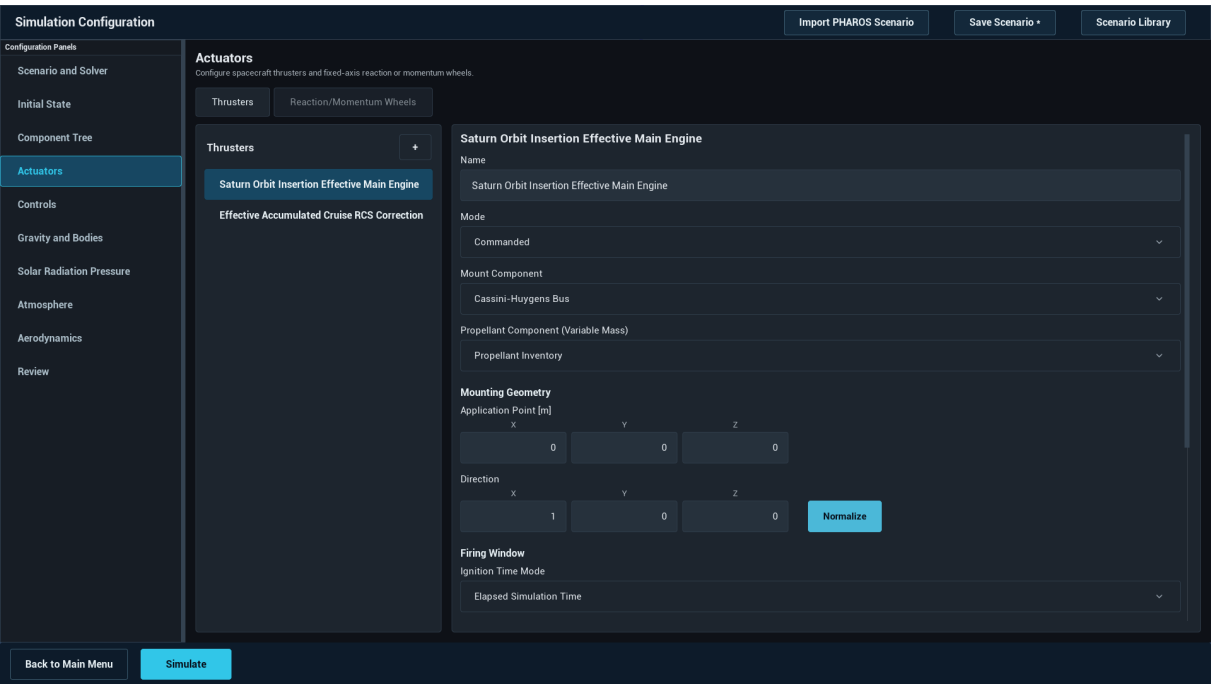


FIGURE E.5 – Actuators panel: thrusters.

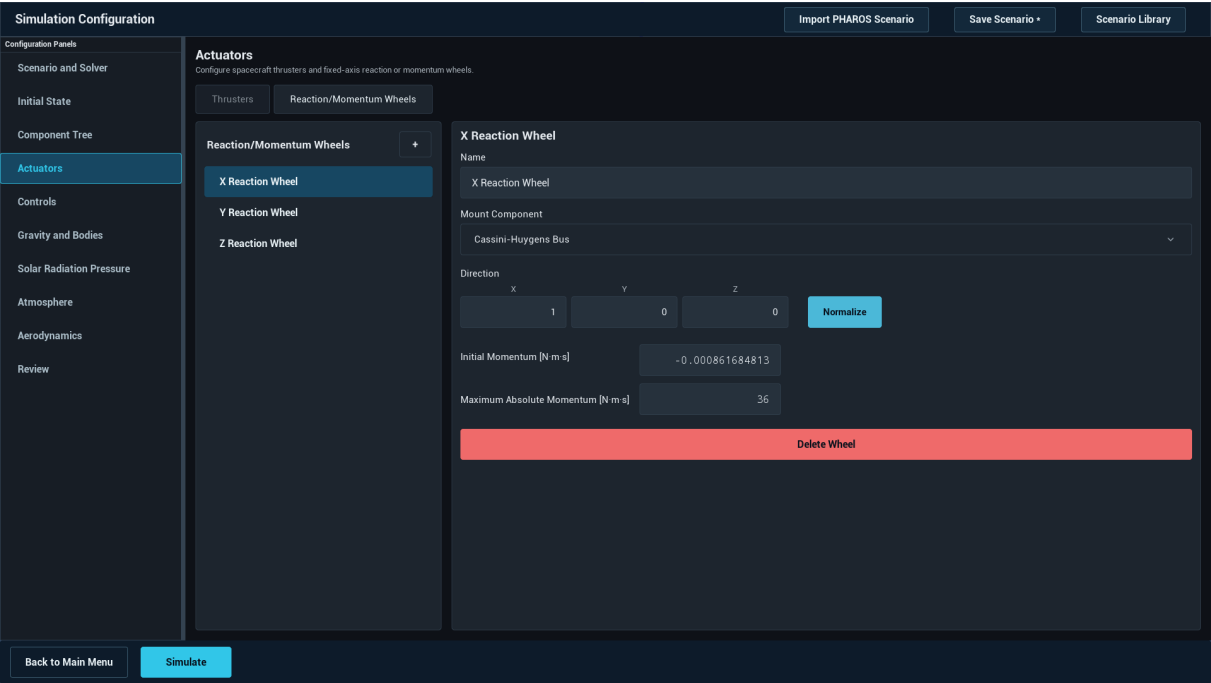


FIGURE E.6 – Actuators panel: reaction and momentum wheels.

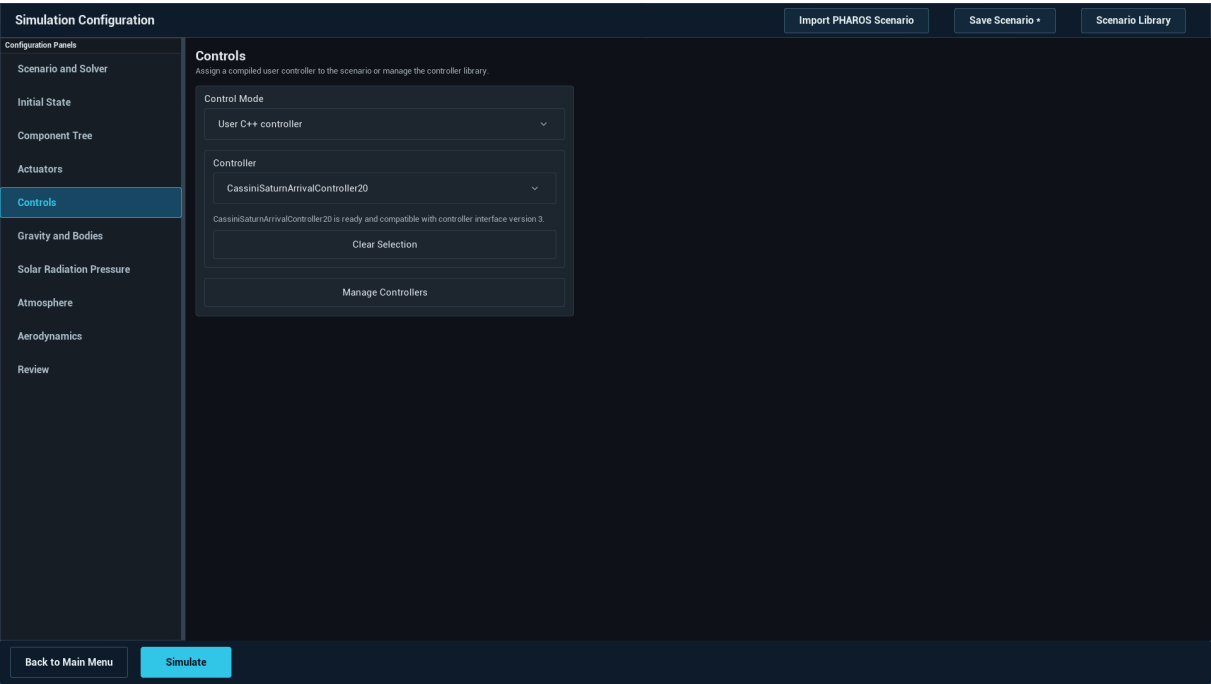


FIGURE E.7 – Controls panel.

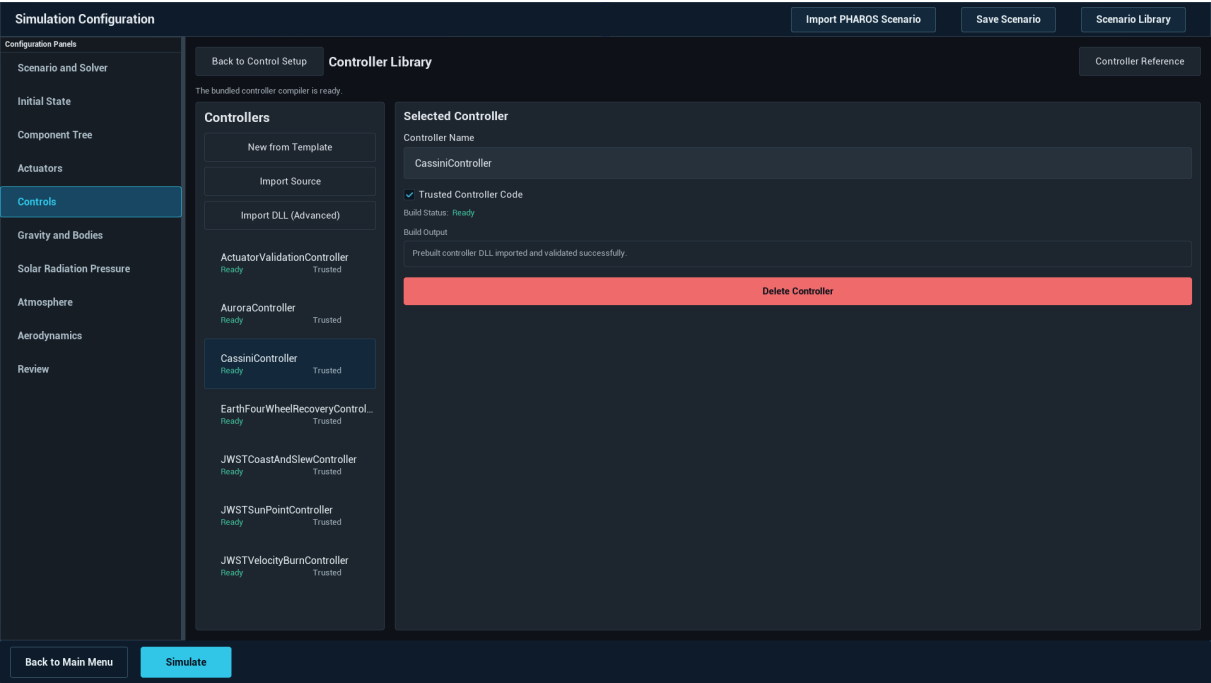


FIGURE E.8 – Controller Library.

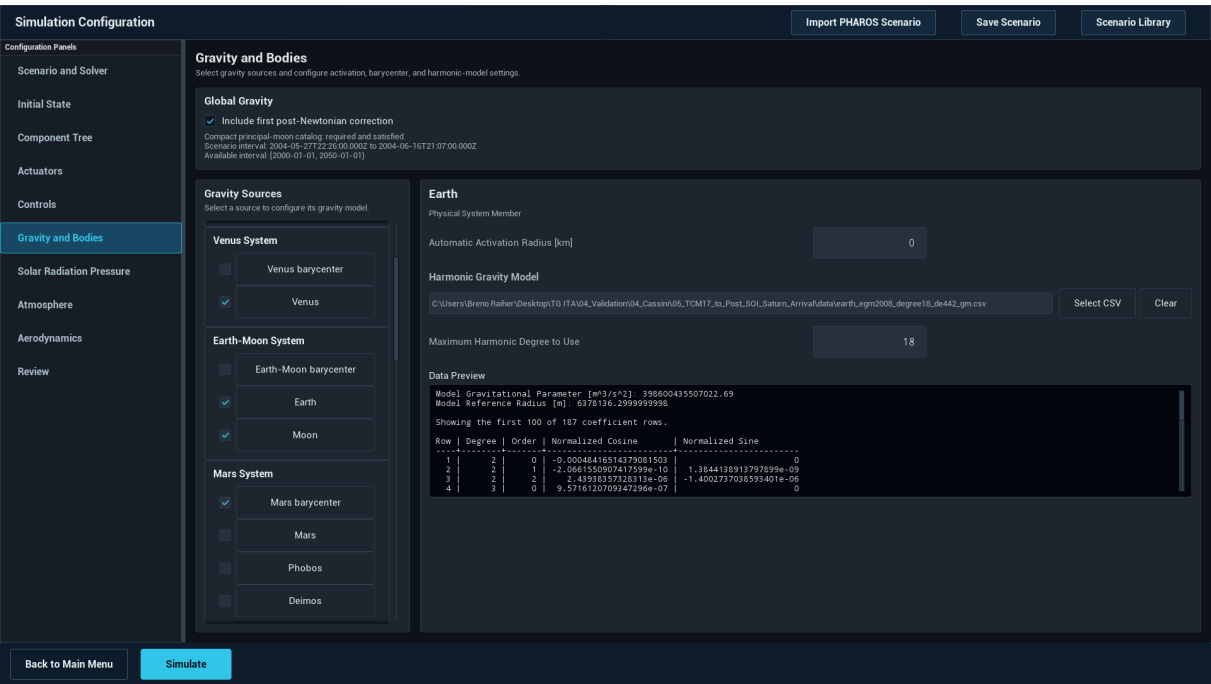


FIGURE E.9 – Gravity and Bodies panel.

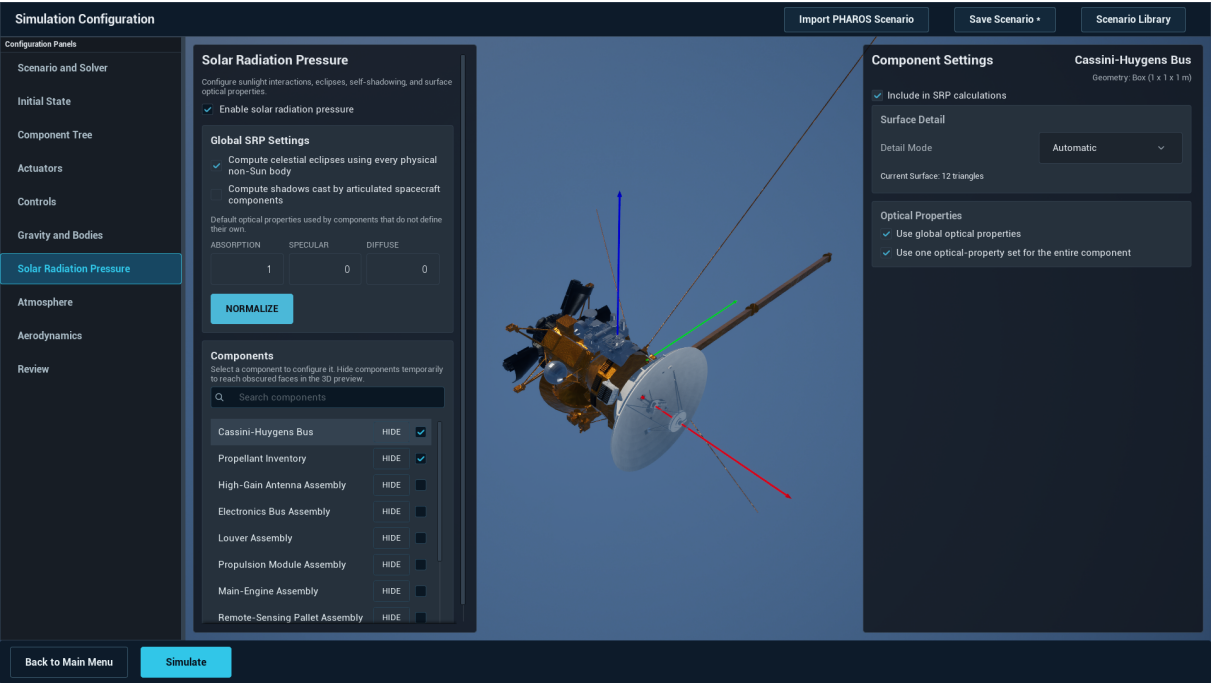


FIGURE E.10 – Solar Radiation Pressure panel.

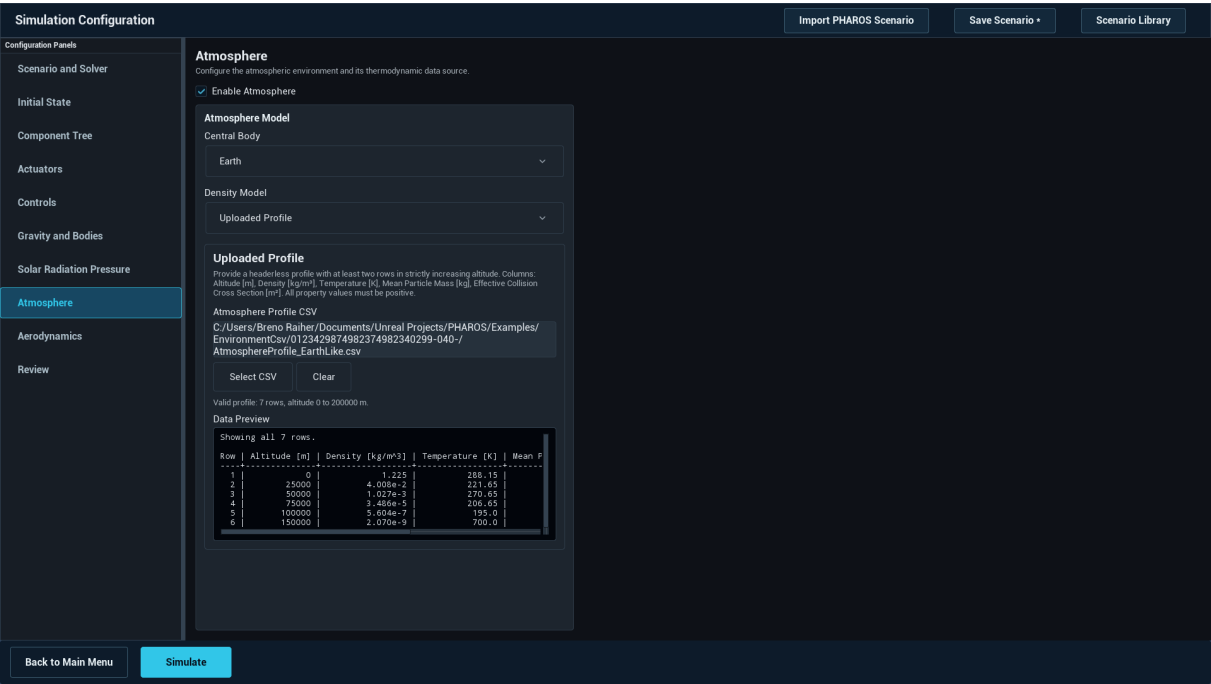


FIGURE E.11 – Atmosphere panel.

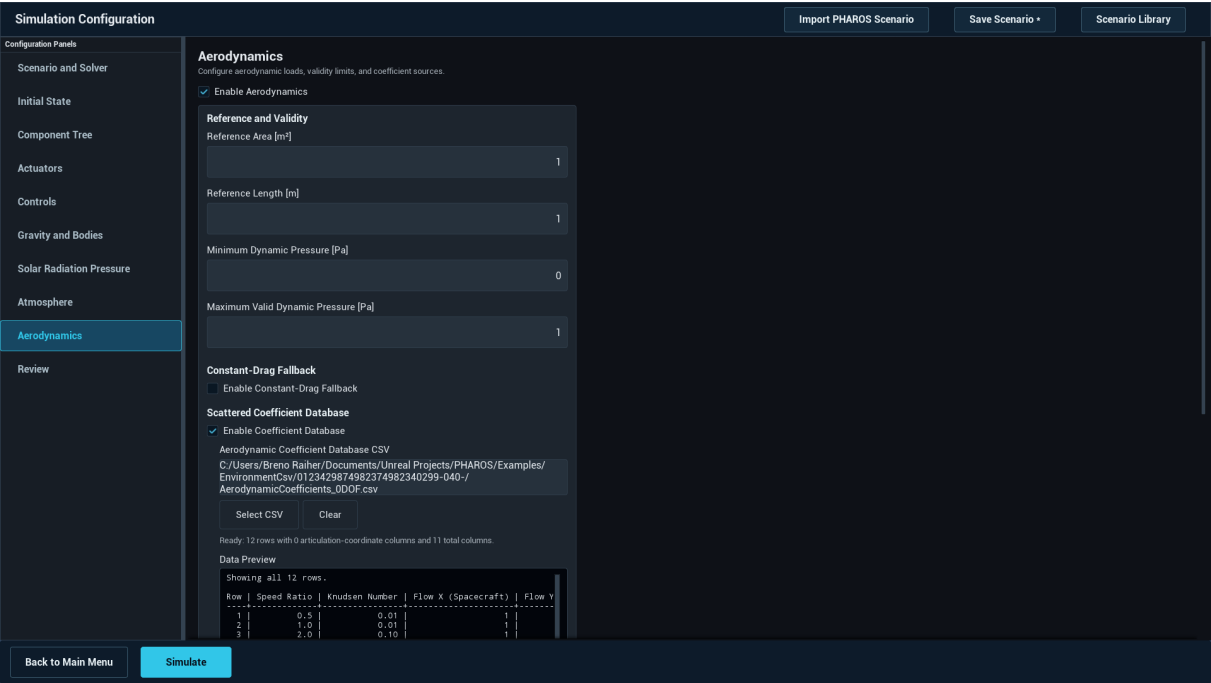


FIGURE E.12 – Aerodynamics panel.

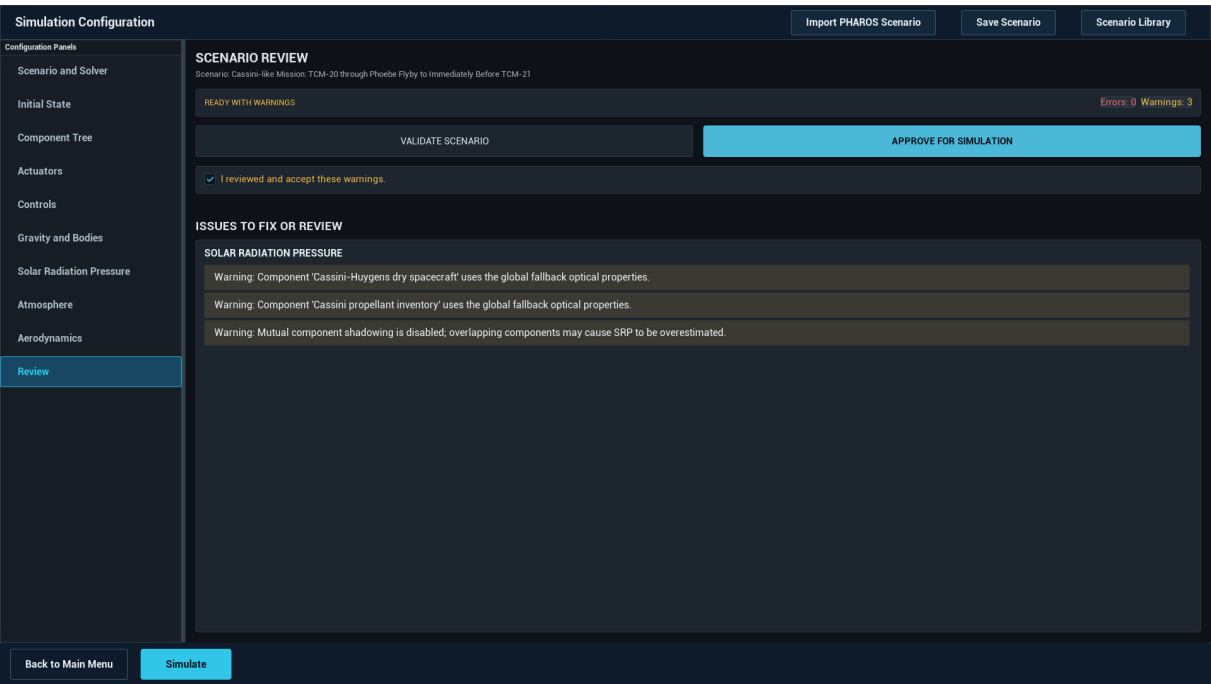


FIGURE E.13 – Review panel.

E.3 Simulation and the Scenario Library

After accepting the configuration in the Review panel, click *Simulate* to start the calculation. PHAROS displays progress and diagnostic messages, and *Cancel* stops the running simulation. Demanding scenarios can take a long time to complete, so the progress bar may advance slowly. To stop an unfinished simulation automatically after a chosen amount of real time, right-click *Simulate* and select *Limit Execution Time*. PHAROS cancels the run when this limit is reached. The limit concerns actual execution time, independently of the simulated duration; it does not accelerate the calculation or ensure completion within that time.

Find run results in %LOCALAPPDATA%\PHAROS\Saved\SimulationResults. Each run has its own folder. Open the folder named after the scenario, then the run subfolder identified by its start timestamp and unique identifier. It contains `scenario.tgscn`, `scenario_solution.csv`, `scenario_summary.txt`, and `run.log`. The CSV holds the recorded numerical results; the summary and log report the execution outcome and diagnostics. For independent analysis, results can be copied from this application-data directory.

The Scenario Library, shown in Figure E.14, provides search and sorting for saved scenarios and indicates whether their results are available. Select a scenario to reopen it for editing or load its result for visualization. The continuation command creates a new scenario whose initial dynamic state comes from the final recorded state of a completed run. The associated TGSCN and CSV files must remain available for visualization and continuation; the library’s .sav file does not replace these result files.

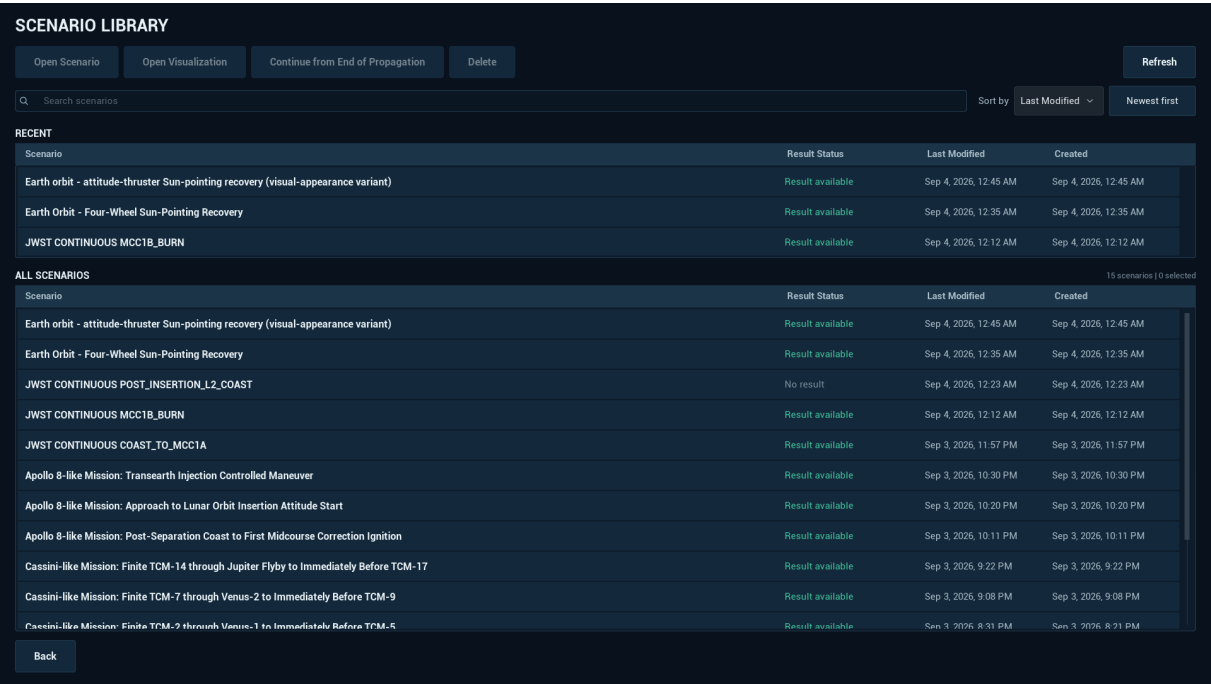


FIGURE E.14 – Scenario Library.

E.4 Standalone Validation and Propagation

The standalone runner uses the same scenario preparation, physical models, and recorder as the graphical application. The following PowerShell examples use the default installation directory; change `$pharosRoot` if another directory was selected. Run the commands from the directory containing `mission.tgscn`, with its referenced input files accessible. The SPICE kernels are separate data files supplied with PHAROS, not embedded in the executable. First validate the inputs without propagating:

```
$pharosRoot = "$env:LOCALAPPDATA\Programs\PHAROS"
$runner = "$pharosRoot\PHAROS\Binaries\Win64\PHAROSScenarioRunner.exe"
$ernels = "$pharosRoot\PHAROS\Content\SPICEKernels"
& $runner ".\mission.tgscn" --validate-only --kernel-dir $ernels
```

Then propagate to a separate output directory:

```
& $runner ".\mission.tgscn" `
  --output ".\results\run_001" --kernel-dir $ernels
```

Validation loads the kernels and any selected controller, reads the scenario, resolves its references, and checks the numerical configuration. It does not integrate or write a solution CSV. The execution-time limit available in the graphical application does not apply to a direct command-line run; a separate process supervisor is needed if such a deadline is required.

The runner prints progress and diagnostics to the terminal and returns one of the codes in Table E.2. For `mission.tgscn`, the output directory receives `mission_solution.csv` and `mission_summary.txt`. The CSV begins with absolute ephemeris time and independent elapsed time, followed by the recorded spacecraft, component, actuator, environment, force, and moment telemetry. Check the exit code and summary before using output from an interrupted or failed run.

TABLE E.2 – Exit codes returned by PHAROS Scenario Runner.

Code	Meaning
0	Validation or propagation succeeded
2	Invalid command-line arguments
3	TGSCN parse failure
4	Controller DLL load or compatibility failure
5	SPICE-kernel load failure
6	Scenario preparation or configuration-validation failure
7	Output directory creation failure
8	Result-file write failure
9	Numerical propagation returned a failed result

E.5 Result Visualization

Opening a completed run presents a spacecraft-centered three-dimensional view driven by the stored samples. The timeline can play, pause, change rate, and seek by mission time, ephemeris time, or UTC. Dockable panels provide:

- spacecraft, component, actuator, force, moment, and closest-body telemetry;
- heliocentric and closest-body three-dimensional trajectory overviews;
- selectable vectors for directions to celestial bodies, state quantities, forces, and moments;
- constellation outlines; and
- two- or three-dimensional plots of compatible recorded columns over a selected time range.

The *Options* button in the visualization header opens controls for *Bodies Emissive* and *Spacecraft Emissive*, camera field of view and mouse sensitivity, and the local trajectory's displayed time range, color, and reference frame (*ICRF* or *Closest Body*). Enable the emissive options to illuminate otherwise dark sides of celestial bodies or the spacecraft, making their details easier to inspect during playback. Figures E.15–E.17 show representative panel arrangements.

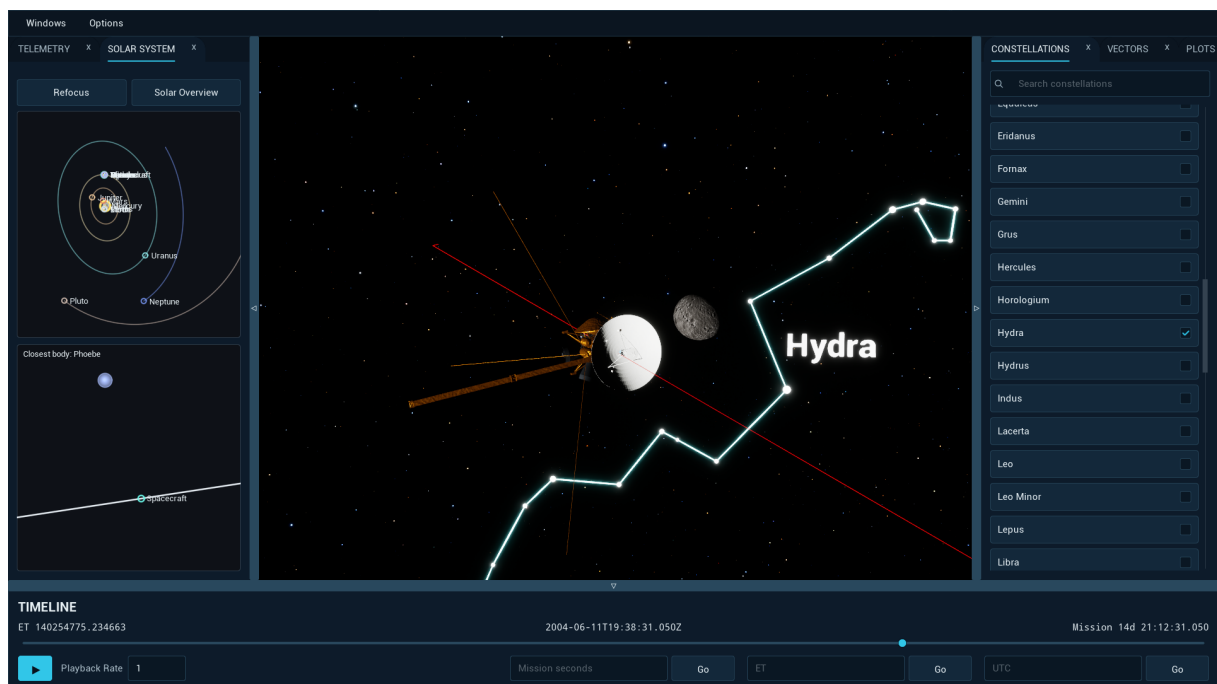


FIGURE E.15 – Visualization with Solar System overviews and constellations.

Trajectory paths span astronomical distances, so local changes in direction can look abrupt in a wide view. The default time range shows the full recorded trajectory. When a scenario includes several revolutions around the same body, overlapping passages can resemble branches. Narrow the displayed range to isolate the revolution or encounter of interest, and zoom in to inspect its geometry.

In *ICRF*, the path combines spacecraft positions from different epochs while each celestial body is shown at the current playback time. The line can therefore appear to pass through a body's current position even when the spacecraft did not collide with it. Select *Closest Body* to inspect the trajectory relative to that body: this removes the body's translation at each sampled instant and resolves apparent crossings caused by that translation. Playback still relies on recorded samples, so a sufficiently short recording interval is needed to resolve brief events.



FIGURE E.16 – Visualization with telemetry and plots.

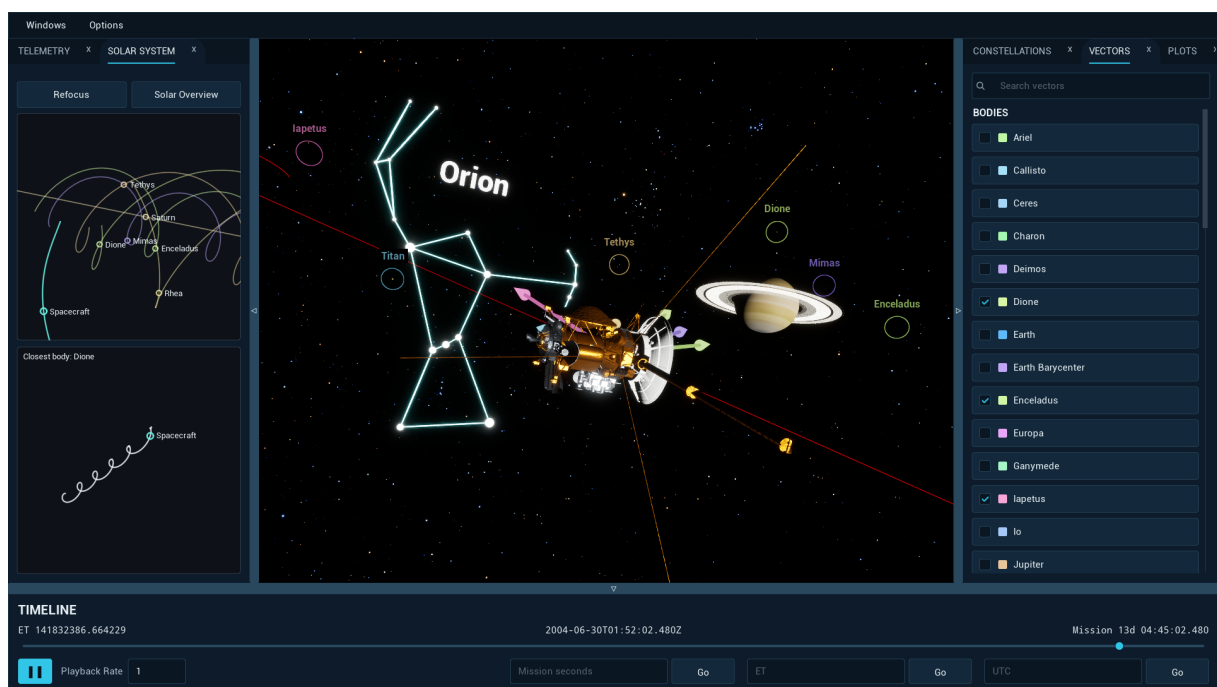


FIGURE E.17 – Visualization with direction vectors.